

SAT UZBEKISTAN

1

Mark for review

Abc

$$5z + 10 = 35$$

Which equation has the same solution as the given equation?

A $5z = 45$

B $5z = 25$

C $5z = 15$

D $5z = 40$

A

B

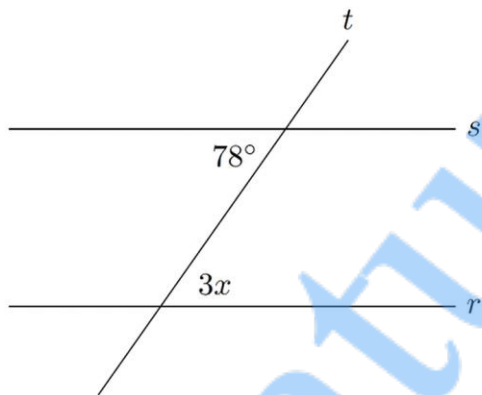
C

D

2

Mark for review

Abc



Note: Figure not drawn to scale

In the figure, line r is parallel to line s . What is the value of x ?

A 12°

B 26°

C 78°

D 102°

A

B

C

D

3

 Mark for review

Each value in the data set shown represents the number of minutes it took a group of runners to complete a race.

50, 42, 40, 53, 48, 35, 45

What is the median time, in minutes, to complete the race?

A) 45 B) 53 C) 48 D) 42

4

 Mark for review

Abc

$$3p + 5f = 89$$

The given equation describes the relationship between the number of paintings, p , and the number of frames, f , that can be stored in an art gallery on a given day. If the gallery stores 10 frames on a given day, how many paintings can it store on this day?

A

10

★

B

13

B

C

30

€

D

50

B

5

 Mark for review

Abe

A cone has a base diameter and a height of 54 units each. What is the volume, in cubic units, of the cone?

A $13,122\pi$

A

B $39,366\pi$

B

C $52,488\pi$

C

D $157,464\pi$

D

6

 Mark for review

9 is $p\%$ of 40. What is the value of p ?

A) 20.5 B) 22.5 C) 24.5 D) 26.5

7

 Mark for review

Abe

A model predicts that the amount of waste produced by a city was 5,000 tons in 2000. The model also predicts that each year for the next 8 years, the amount of waste w increased by 2% of the previous year's amount. Which equation best represents this model, where x is the number of years after 2000, for $x \leq 8$?

A $w = 1.02(5,000)^x$

A

B $w = 0.98(5,000)^x$

B

C $w = 5,000(1.02)^x$

C

D $w = 5,000(0.98)^x$

D

8

 Mark for review

Abe

$$3x^2 + 5x = 42$$

What is the positive solution to the given equation?

A $\frac{5}{2}$

A

B 3

B

C $\frac{14}{3}$

C

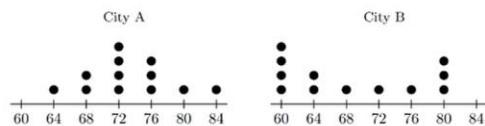
D 14

D

9

 Mark for review

Abc



The dot plots are shown, representing the monthly temperature averages in City A and City B over a year. Which statement best compares the standard deviations of the monthly temperature averages in these two cities?

A The standard deviation of the monthly temperature averages in City A is less than the standard deviation of the monthly temperature averages in City B.

A

B The standard deviation of the monthly temperature averages in City A is equal to the standard deviation of the monthly temperature averages in City B.

B

C The standard deviation of the monthly temperature averages in City A is greater than the standard deviation of the monthly temperature averages in City B.

C

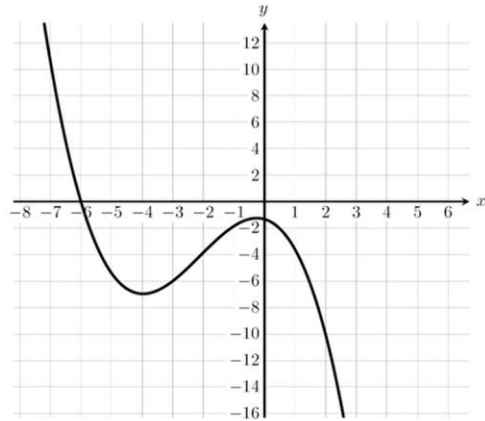
D There is not enough information to compare these standard deviations.

D

10

Mark for review

Abc



The graph of $y = g(x)$ is shown, where the function g is defined by $g(x) = px^3 + qx^2 + rx + s$ and p, q, r , and s are constants. How many distinct values of x does $g(x) = 0$ have?

A One

A

B Two

B

C Three

C

D Four

D

11

Mark for review

The exponential function f is defined by $f(x) = 4 \cdot c^x$, where c is a positive constant. If $f(4) = 5,184$, what is the value of $f(5)$?

A) 31114 B) 31104 C) 32116 D) 32104

12

 Mark for review

Abc

$$y = (x - 3)(x + 5)$$

$$y = 7x - 21$$

Which ordered pair (x, y) is the solution to the given system of equations?

A $(0, 3)$

B $(-5, -24)$

C $(3, 0)$

D $(-5, 0)$

A

B

C

D

13

 Mark for review

Abc

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles B and E are right angles. If $\sin(A) = \frac{180}{181}$, what is the value of $\cos(F)$?

A $\frac{19}{181}$

B $\frac{180}{181}$

C $\frac{181}{19}$

D $\frac{180}{19}$

A

B

C

D

14

 Mark for review

The function $f(t) = 500(3)^{\frac{t}{91}}$ represents the number of fish in a pond t days after an initial observation. How many days does it take for the number of fish in the pond to be nine times the initial number?

A) 91 B) 182 C) 273 D) 364

15

 Mark for review

Abe

$$r = \frac{m}{3n + 5}$$

The given equation relates the distinct positive numbers r , m , and n . Which equation correctly expresses $3n$ in terms of r and m ?

A $3n = \frac{m}{r} - 5$

A

B $3n = rm - 5$

B

C $3n = 5r - m$

C

D $3n = \frac{r - 5}{m}$

D

16

 Mark for review

Line m is defined by $5x - 3y = 9$. Line n is perpendicular to line m in the xy -plane. What is the slope of line n ?

A) -3/5 B) -5/3 C) 3/5 D) 5/3

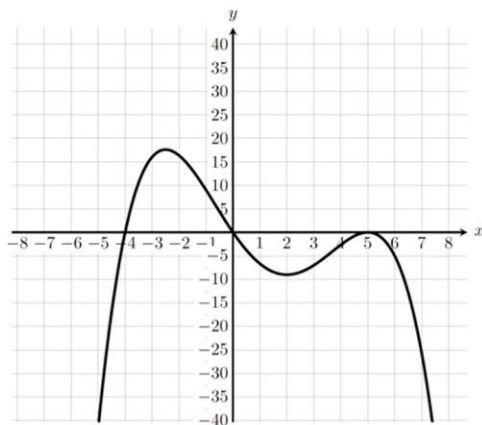
17  Mark for review

Point A is the center of a circle. The measure of arc BC on this circle is 107° . What is the measure, in degrees, of its associated angle BAC ?

A) 107 B) 73 C) 117 D) 63

18  Mark for review

Abc



Which of the following could be the equation of the graph shown in the xy -plane?

A $y = -\frac{1}{12}x(x-5)^2(x+4)$

A

B $y = -\frac{1}{12}x^2(x-5)(x+4)$

B

C $y = -\frac{1}{12}x(x-5)(x+4)$

C

D $y = -\frac{1}{12}x(x-5)(x+4)^2$

D

19

 Mark for review

Abc

For $x > 0$, the function f is defined as follows:

$f(x)$ is obtained by halving each value of x .

Which of the following could describe this function?

A Decreasing exponential

A

B Decreasing linear

B

C Increasing exponential

C

D Increasing linear

D

20

 Mark for review

Abc

$$g(x) = 2x^2 - 52x + 317$$

The function h is defined by $h(x) = g(x - 4)$. For what value of x does $h(x)$ reach its minimum?

A 9

A

B 13

B

C 17

C

D 317

D

21

Mark for review

Abc

Music Preferences

Rock	147
Pop	109
Classical	51

The table shows the results of a survey of people's music preferences. A total of 307 people were surveyed. If 1,543 people attend a concert featuring various genres, how many would be expected to prefer rock music based on the survey results?

A 256

A

B 548

B

C 739

C

D 1,543

D

22

Mark for review

Abc

$$y = ax^2 + 3x + 5$$

$$y = -x - 3$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

A -3

A

B -5

B

C 5

C

D -4

D

Module 2

1

🔖 Mark for review

Abc

$$4x - 3y = -5$$

$$y = 2x - 7$$

The solution to the given system of equations is (x, y) . What is the value of y ?

A -7

B 13

C 19

D 25

2

🔖 Mark for review

Abc

Which expression is equivalent to $11a^2b^2 - 3a^2b^2$?

A $8a^2b^2$

B $14a^2b^2$

C $14a^4b^4$

D $8a^4b^4$

3

 Mark for review

Abc

$$36x - 27 = 3(12x - 3)$$

How many solutions does the given equation have?

☐ A Exactly one☐ B Exactly two☐ C Infinitely many☐ D Zero

4

 Mark for review

Abc

A new community center received votes on a city ballot. There are four times as many people who voted in favor of the center as those who voted against it. A blog post mentioned that 12,000 more people voted in favor of the center than voted against it. Based on this information, how many people voted against the community center?

☐ A 3,000☐ B 4,000☐ C 12,000☐ D 16,000

5

 Mark for review

Abc

A coffee shop sells coffee and the function $h(x) = -3x + 240$ represents the number of cups of coffee remaining after selling x cups. What is the best interpretation of the slope of the graph of $y = h(x)$ in the xy -plane in this context?

- ☐ A The coffee shop sells 3 cups of coffee per hour.
- ☐ B The coffee shop initially had 3 cups of coffee available.
- ☐ C The coffee shop initially had 240 cups of coffee available.
- ☐ D The coffee shop sells 240 cups of coffee per hour.

6

 Mark for review

An angle has a measure of $\frac{7\pi}{9}$ radians. What is the measure of the angle, in degrees?

-
- A) 140 B) 160 C) 200 D) 220

7

 Mark for review

Abc

$$y \geq 4x - 2$$

$$y \leq -x + 5$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

A) $(0, 2)$

B) $(1, -2)$

C) $(2, 6)$

D) $(-1, 7)$

8

 Mark for review

A right triangle has legs with lengths of 24 centimeters and 28 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $k\sqrt{d}$, where k and d are integers, what is the value of d ?

A) 65 B) 85 C) 75 D) 55

9

Mark for review

Abc

Value	Data set X Frequency	Data set Y Frequency
23	3	7
26	6	5
29	8	7
32	5	4
35	3	2

Data set X and data set Y each consist of 25 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the medians of the two data sets?

- ☐ A The median of data set X is greater than the median of data set Y .
- ☐ B The median of data set X is less than the median of data set Y .
- ☐ C The median of data set X is equal to the median of data set Y .
- ☐ D There is not enough information to compare the medians of the data sets.

10

Mark for review

Abc

Triangle JKL is similar to triangle UVW such that J, K , and L correspond to U, V , and W , respectively. The measure of $\angle L$ is 60° and $JK = \frac{1}{4}UV$. What is the measure of $\angle W$?

- ☐ A 15°
- ☐ B 30°
- ☐ C 45°
- ☐ D 60°

11

 Mark for review

Abc

Ella made 24 liters of lemonade. She then filled a small bottles and b large bottles with all the lemonade she made. The equation $2a + 4b = 24$ represents this situation. Which is the best interpretation of $4b$ in this context?

- ☐ A The number of large bottles Ella filled
- ☐ B The number of small bottles Ella filled
- ☐ C The total number of liters of lemonade in the large bottles
- ☐ D The total number of liters of lemonade in the small bottles

12

 Mark for review

$$2x^2 - 5x - 12 = 3x(x - 4)$$

What is the sum of the solutions to the given equation?

- A) 5 B) 14 C) 7 D) 12

13

 Mark for review

Abe

The function $k(x) = 2(x - 3)^2 + 10$ represents the height, in meters, of a balloon $k(x)$, x minutes after it was released, where $0 \leq x \leq 20$. Which of the following is the best interpretation of the vertex of the graph of $y = k(x)$ in the xy -plane?

- ☐ A The balloon's minimum height was 10 meters.
- ☐ B The balloon's minimum height was 3 meters.
- ☐ C The balloon's height was 10 meters when it was released.
- ☐ D The balloon's height was 3 meters when it was released.

14

 Mark for review

Abe

$$P(x) = \frac{3}{2}(x - 100) + 200$$

The function P gives the price, in dollars, of an item when the supply is x units. If the supply increased by 12 units, by how much did the price increase, in dollars?

- ☐ A 3
- ☐ B 6
- ☐ C 9
- ☐ D 18

15

 Mark for review

x	y
$p - 2$	10
$p + 4$	22

The table gives the coordinates of two points on a line in the xy -plane. The x -intercept of the line is $(q, p - 10)$ where p and q are constants. What is the value of q ?

A) 2 B) 3 C) 5 D) 7

16

 Mark for review

One of the factors of $6x^2 - 48x + 72$ is $x + b$, where b is a negative constant. What is the smallest possible value of b ?

A) -3 B) -5 C) -6 D) -9

17

 Mark for review

Abe

The function g is defined by $g(x) = 5x + 75$. What is the x -intercept of the graph of $y = g(x)$ in the xy -plane?

☐ A (0, 15)☐ B (5, 0)☐ C (-15, 0)☐ D (-5, 0)

18

 Mark for review

Abe

A rectangular garden has an area of 9,600 square meters. What is the area, in square centimeters, of this garden? (1 meter = 100 centimeters)

☐ A 0.96☐ B 0.0096☐ C 96,000,000☐ D 9,600,000,000

19

 Mark for review

Abc

A pool has a volume of v cubic meters and a depth of d meters. If the pool is in the shape of a rectangular prism with a length of l meters, which equation represents the width, w , of the pool in meters?

A $w = \frac{v}{ld}$

B $w = \frac{v}{l+d}$

C $w = v - (l + d)$

D $w = vl + d$

20

 Mark for review

$$26x + 92y = 82y + 70$$

$$25y = \frac{1}{14} - kx$$

In the given system of equations, k is a constant. If the system has no solution, what is the value of k ?

A) 65 B) 130 C) 45 D) 55

21

 Mark for review

Abc

$$\frac{(3x)^2}{\sqrt{x^2 - b^2}} = \frac{(3b)^2}{\sqrt{x^2 - b^2}} + 36$$

In the given equation, b is a positive constant. Which of the following is one of the solutions to the given equation?

A $-\sqrt{4^2 + b^2}$

B $-\sqrt{6^2 + b^2}$

C $-\sqrt{12^2 + b^2}$

D $-\sqrt{36^2 + b^2}$

22

 Mark for review

Abc

$$k(x) = rx^2 + 8x + t$$

In the given quadratic function, r and t are constants. The graph of $y = k(x)$ in the xy -plane is a parabola that opens downward and has a vertex at the point (u, v) , where u and v are constants. If $v > 0$ and $k(-1) = k(3) > 0$, which of the following must be true?

- I. $t > 0$
II. $r < -8$

A I only

B II only

C I and II

D Neither I nor II